Recursion - 1

Recusive code → ① Defining function
② Identify the Base Case (Stop Condition)
③ Recursive logic → sum(N) = N + sum(N-1)

- Time - Complexity = O (Number of function calls * Time per function call)

  TC = O(1)

  fact(3) → fact(2) → fact(1)

  T.C for factorial problem

  → T.C = O(N * 1) = O(N)

- Space - Complexity = O (Maximum depth of recursion tree / stackspace * Space per function call)

---

find factorial

```
↳ public class solution{
      public int solve (int A){
          if (A <= 1) return 1;        → Base Condition
          return A * solve (A-1);      → Recursive logic
      }
  }
```

Print 1 to A (in increasing order)

```
public class Solution{
    public void solve (int A){
        Print(A);
        Sopln();
    }
private void print (int i){
    if (i==0) return
    Print(i-1);        → recursive called first
    Sop( i+" ");       → then print
}}
```

Print A to 1 (in decreasing order)

```
public class Solution
    Public void solve (int A){
        Print(A);
        sopln();
    }
Private void print (int i){
    if (i==0) return;
    Sop( i+" ");
    Print(i-1);
}}
```